

EXPERIMENT NO. : 3

DATE: 27/08/2026

1. AIM

To design and verify dynamic D flip-flops in Cadence Virtuoso using a master-slave configuration with (1) using transmission-gate (2) using C²MOS and to calculate the setup time, clock-to-Q delay, and maximum operating frequency for 1–1 and 0–0 overlapping conditions.

2. TOOL

Cadence Virtuoso

3. THEORY

A dynamic D flip-flop stores the input data and transfers it to the output according to the clock transition. In a master-slave configuration, two storage stages operate using complementary clock phases. The master stage captures the input data, while the slave stage transfers the stored data to the output, providing edge-triggered operation.

In the transmission-gate-based implementation, a D latch is first designed using a CMOS transmission gate. Two such latches are then connected in a master-slave configuration with complementary clock signals to form the D flip-flop. In the C²MOS-based implementation, the master-slave D flip-flop is directly designed using clock-controlled CMOS stages.

The setup time (t_{setup}) is the minimum time for which the input D must remain stable before the active clock edge for correct data capture. The clock-to-Q delay (t_{C-Q}) is the time interval between the active clock edge and the corresponding transition at the output Q.

For reliable operation, the minimum clock period is given by $T_{min} = t_{setup} + t_{C-Q}$. Hence, the maximum operating frequency is $f_{max} = 1/T_{min}$. These timing parameters are determined for both 1–1 and 0–0 overlapping conditions using transient simulation in Cadence Virtuoso.

The designed D flip-flops are simulated using 180 nm CMOS technology, and the obtained waveforms are used to calculate the setup time, clock-to-Q delay, minimum clock period, and maximum operating frequency.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

Table 4.2: Result comparison (pre layout vs post layout) of Transmission Gate based dff

parameter	pre layout	post layout	unit	error
t_{setup}	41.74	73.19	ns	+75.3474%
t_{C-Q}	417.36	867.08	ns	+107.7535%
t_{min}	459.1	940.27	ps	+104.8072%
f_{max}	2.1782	1.0635	GHz	-51.1753%

Table 4.3: Result comparison (pre layout vs post layout) of C2MOS based dff

parameter	pre layout	post layout	unit	error
t_{setup}	41.063	41.063	ns	0%
t_{C-Q}	95.36	152.83	ns	+60.2664%
t_{min}	136.423	193.893	ps	+42.1263%
f_{max}	7.3301	5.1575	GHz	-29.6394%

5. CALCULATION FOR TRANSMISSION GATE BASED D FLIP FLOP

$$t_{setup(pre)} = (80.0015 - 79.95976) * 1000 = 41.74ps$$

$$t_{setup(post)} = (80.0015 - 79.92831) * 1000 = 73.19ps$$

$$t_{C-Q(pre)} = (80.41886 - 80.0015) * 1000 = 417.36ps$$

$$t_{C-Q(post)} = (80.86858 - 80.0015) * 1000 = 867.08ps$$

$$T_{min(pre)} = t_{setup(pre)} + t_{C-Q(pre)} = 41.74 + 417.36 = 459.10ps$$

$$T_{min(post)} = t_{setup(post)} + t_{C-Q(post)} = 73.19 + 867.08 = 940.27ps$$

$$f_{max(pre)} = 1/T_{min(pre)} = 1/459.10 = 2.178GHz$$

$$f_{max(post)} = 1/T_{min(post)} = 1/940.27 = 1.063GHz$$

6. CALCULATION FOR C2MOS BASED D FLIP FLOP

$$t_{setup(pre)} = (10.0025 - 9.961437) * 1000 = 41.063ps$$

$$t_{setup(post)} = (10.0025 - 9.961437) * 1000 = 41.063ps$$

$$t_{C-Q(pre)} = (10.09786 - 10.0025) * 1000 = 95.36ps$$

$$t_{C-Q(post)} = (10.15533 - 10.0025) * 1000 = 152.83ps$$

$$T_{min(pre)} = t_{setup(pre)} + t_{C-Q(pre)} = 41.063 + 95.36 = 136.423ps$$

$$T_{min(post)} = t_{setup(post)} + t_{C-Q(post)} = 41.063 + 152.83 = 193.893ps$$

$$f_{max(pre)} = 1/T_{min(pre)} = 1/136.423 = 7.330GHz$$

$$f_{max(post)} = 1/T_{min(post)} = 1/193.893 = 5.157GHz$$

7. CIRCUIT DIAGRAMS & WAVEFORMS

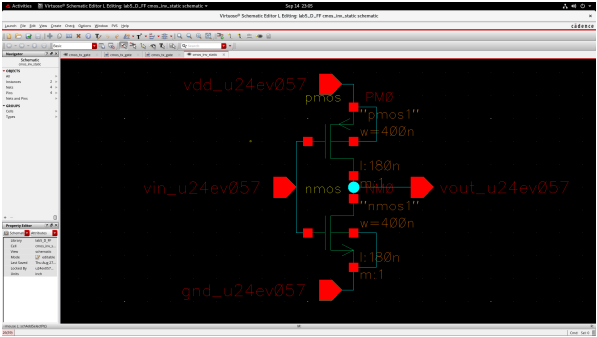


Figure 7.1: Inv Schematic

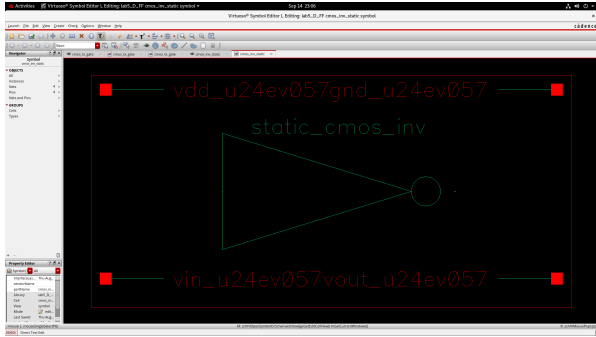


Figure 7.2: Inv Symbol

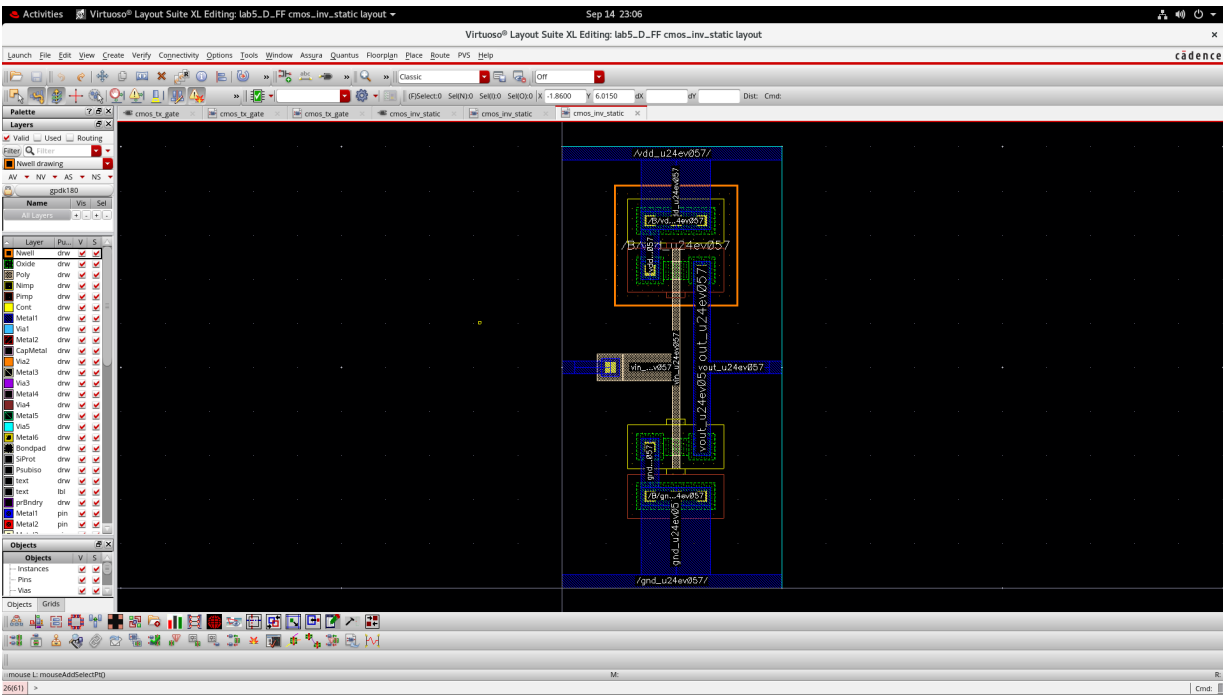


Figure 7.3: Inv Layout

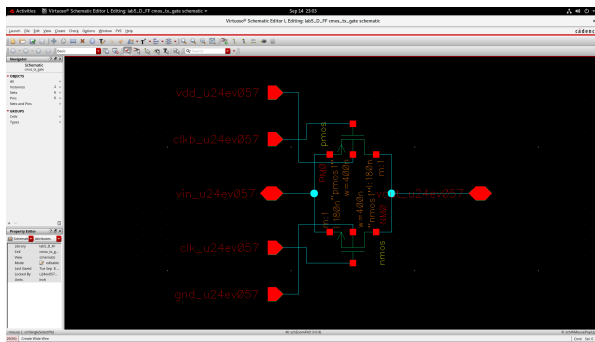


Figure 7.4: Tx Gate Schematic

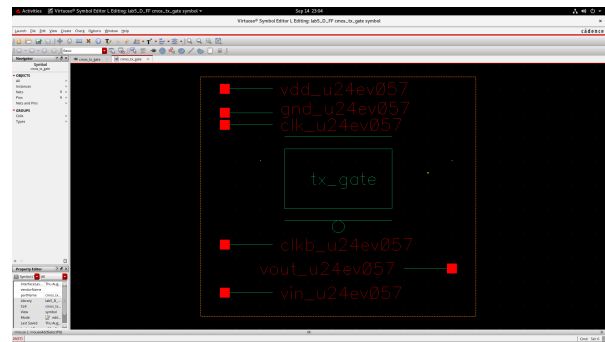


Figure 7.5: Tx Gate Symbol

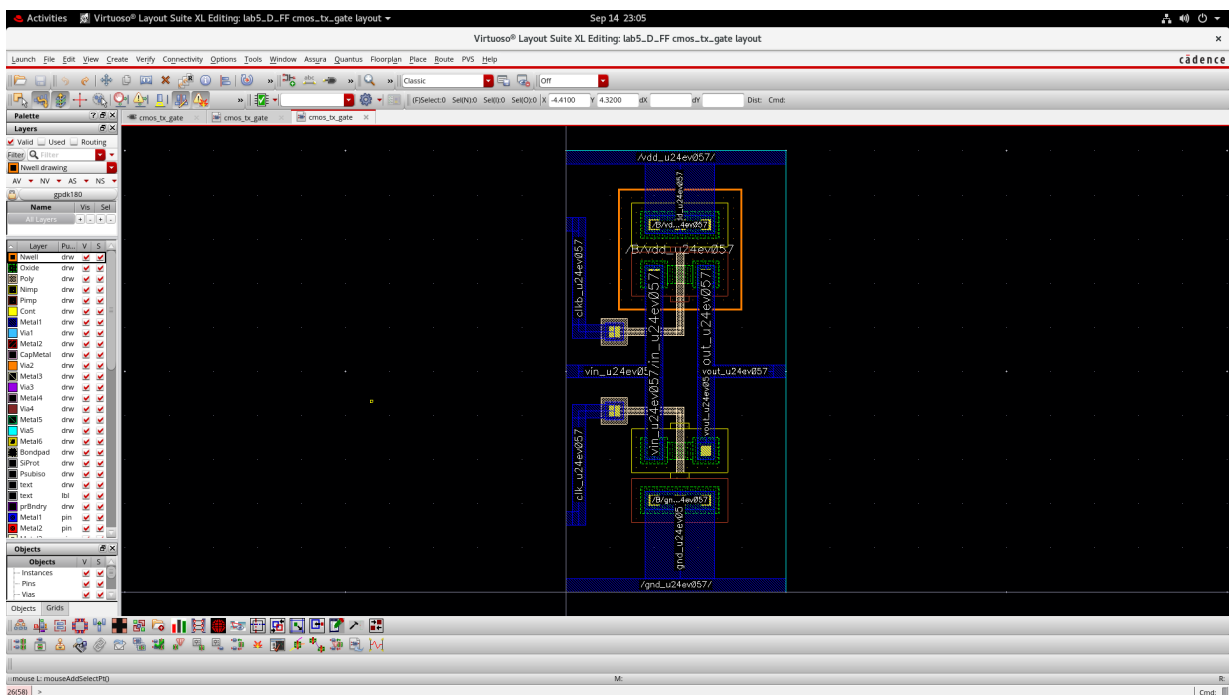


Figure 7.6: Tx Gate Layout

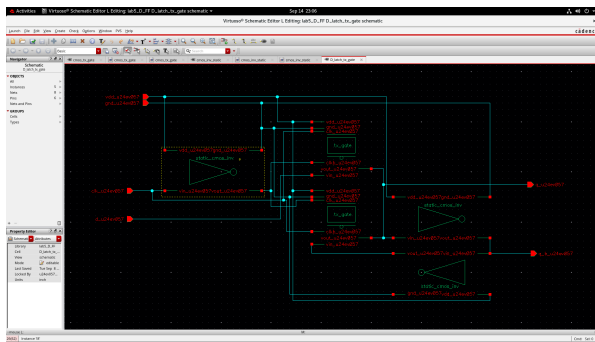


Figure 7.7: Tx Dlatch Schematic

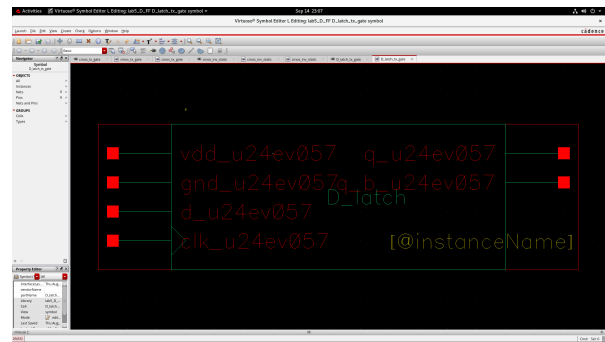


Figure 7.8: Tx Dlatch Symbol

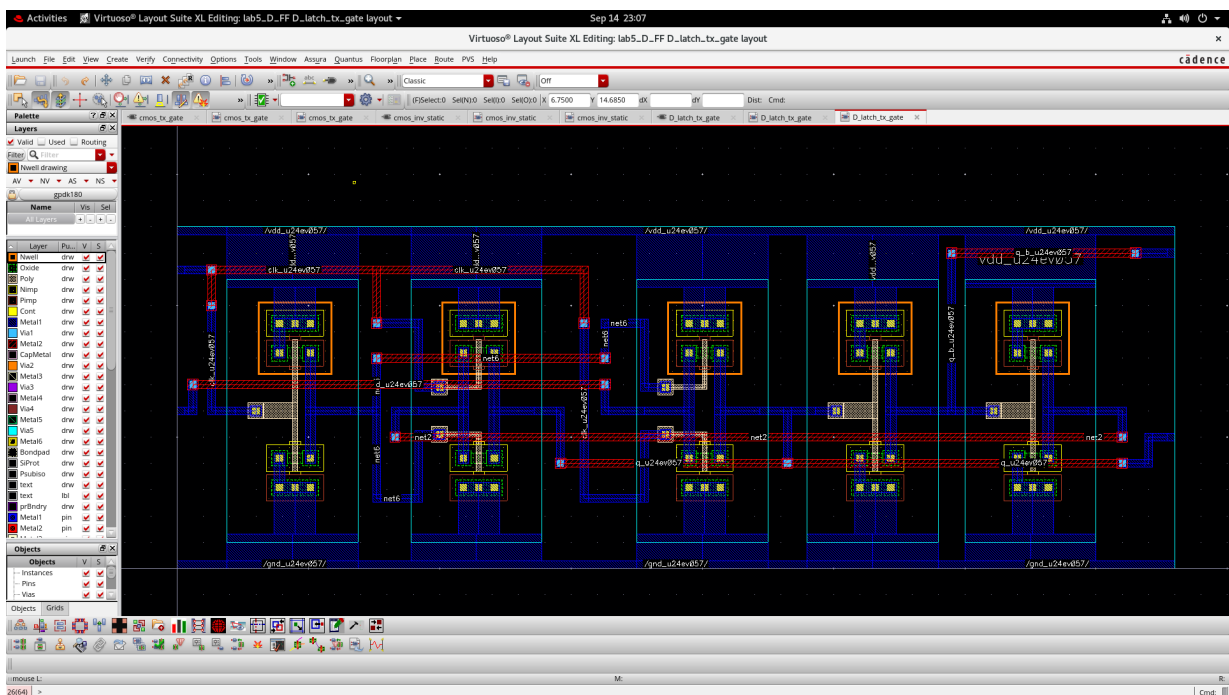


Figure 7.9: Tx Dlatch Layout

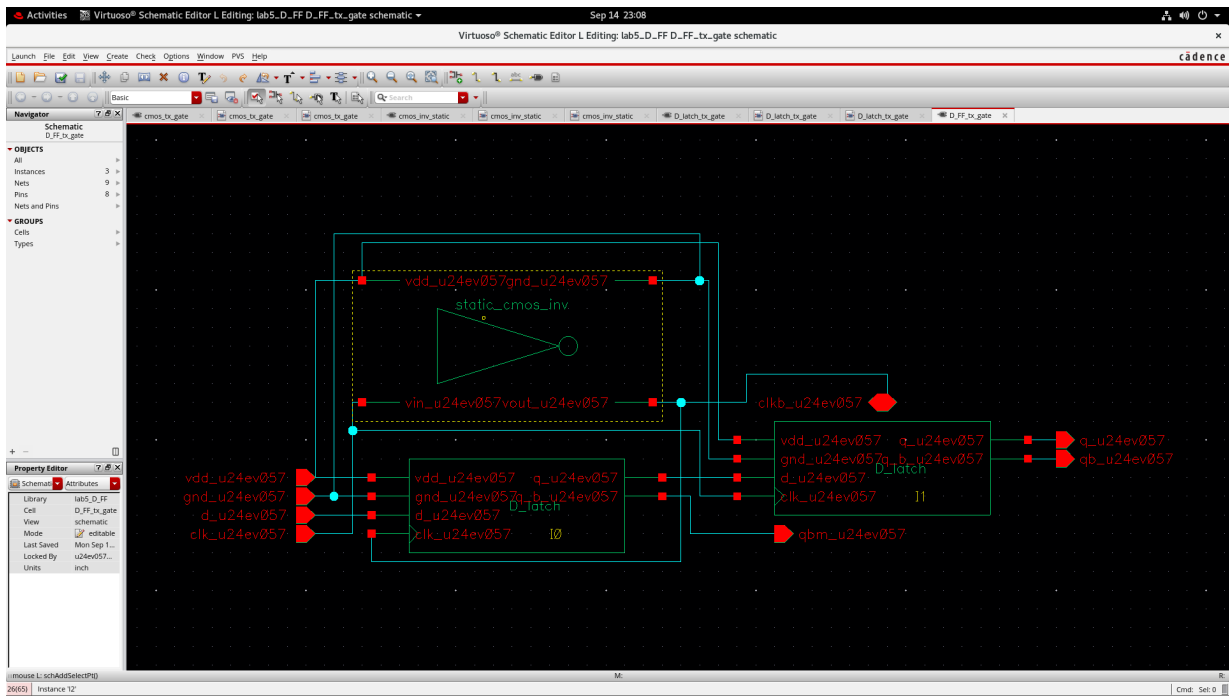


Figure 7.10: Tx Dff Schematic

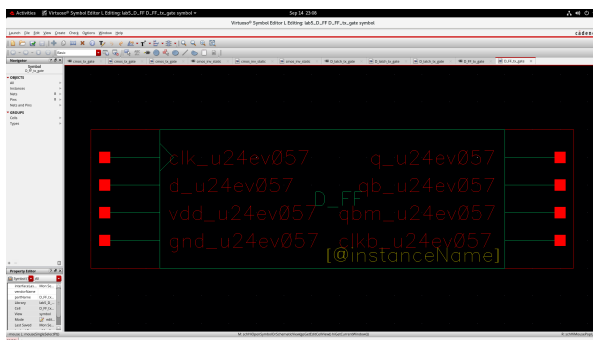


Figure 7.11: Tx Dff Symbol

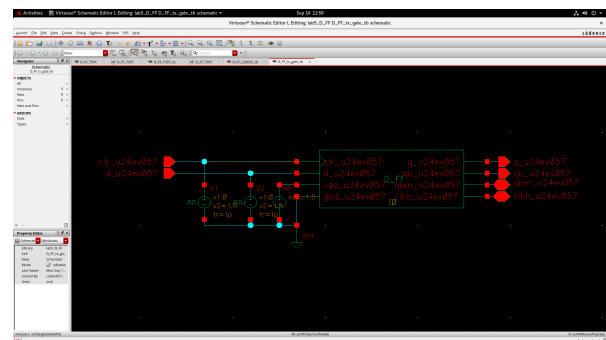


Figure 7.12: Tx Dff Tb

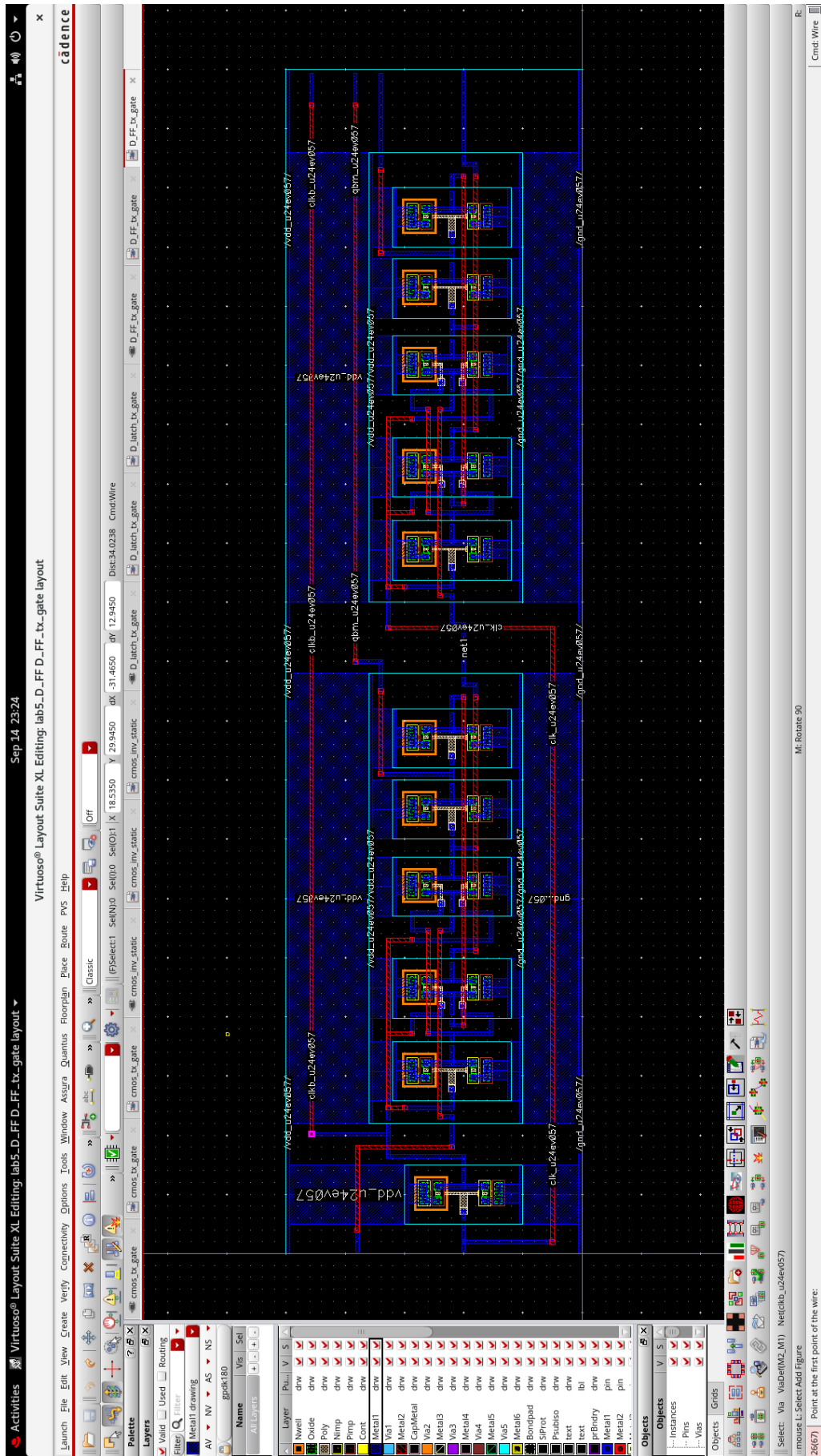


Figure 7.13: Tx Dff Layout

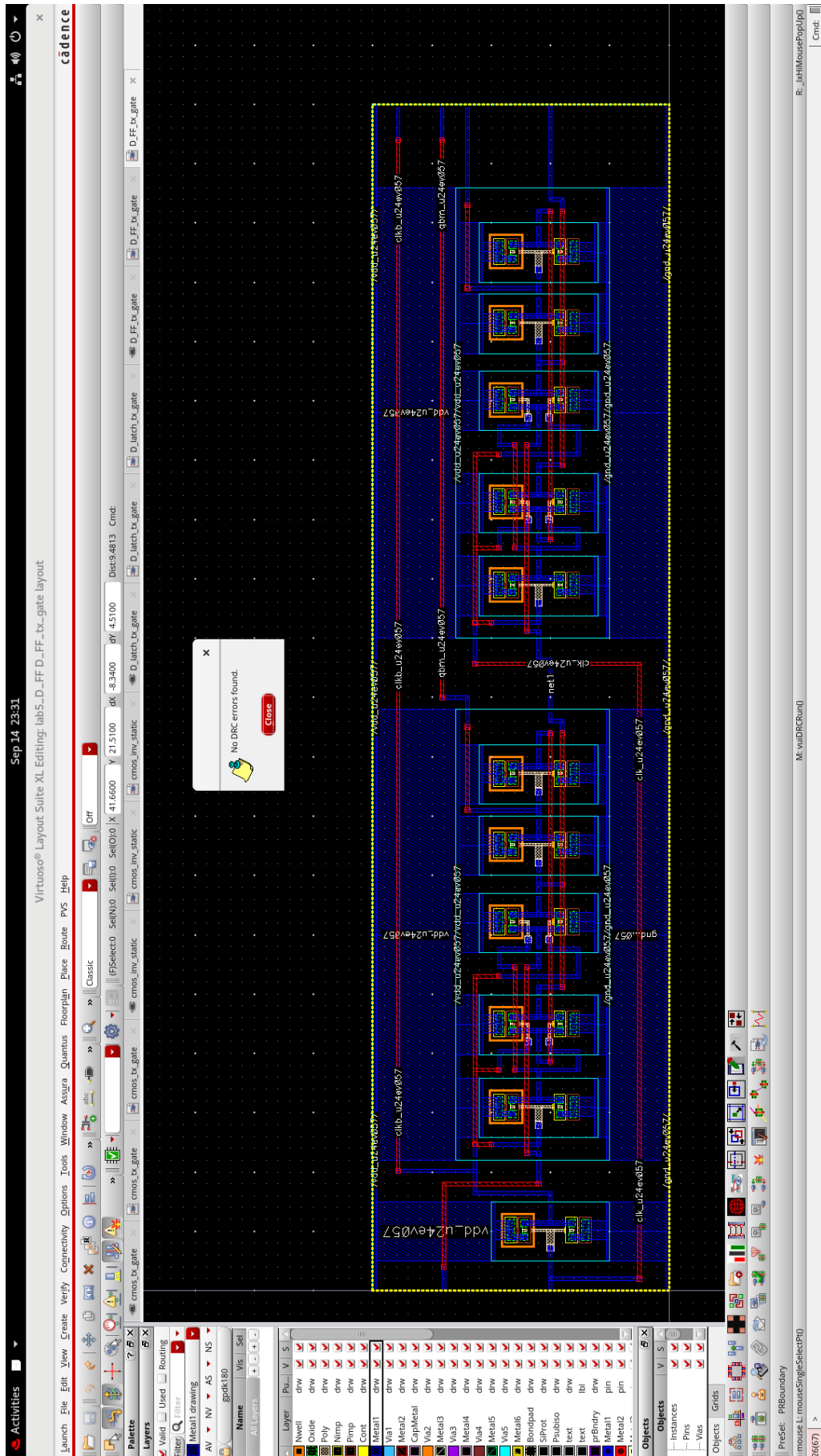


Figure 7.14: Tx Dff Drc

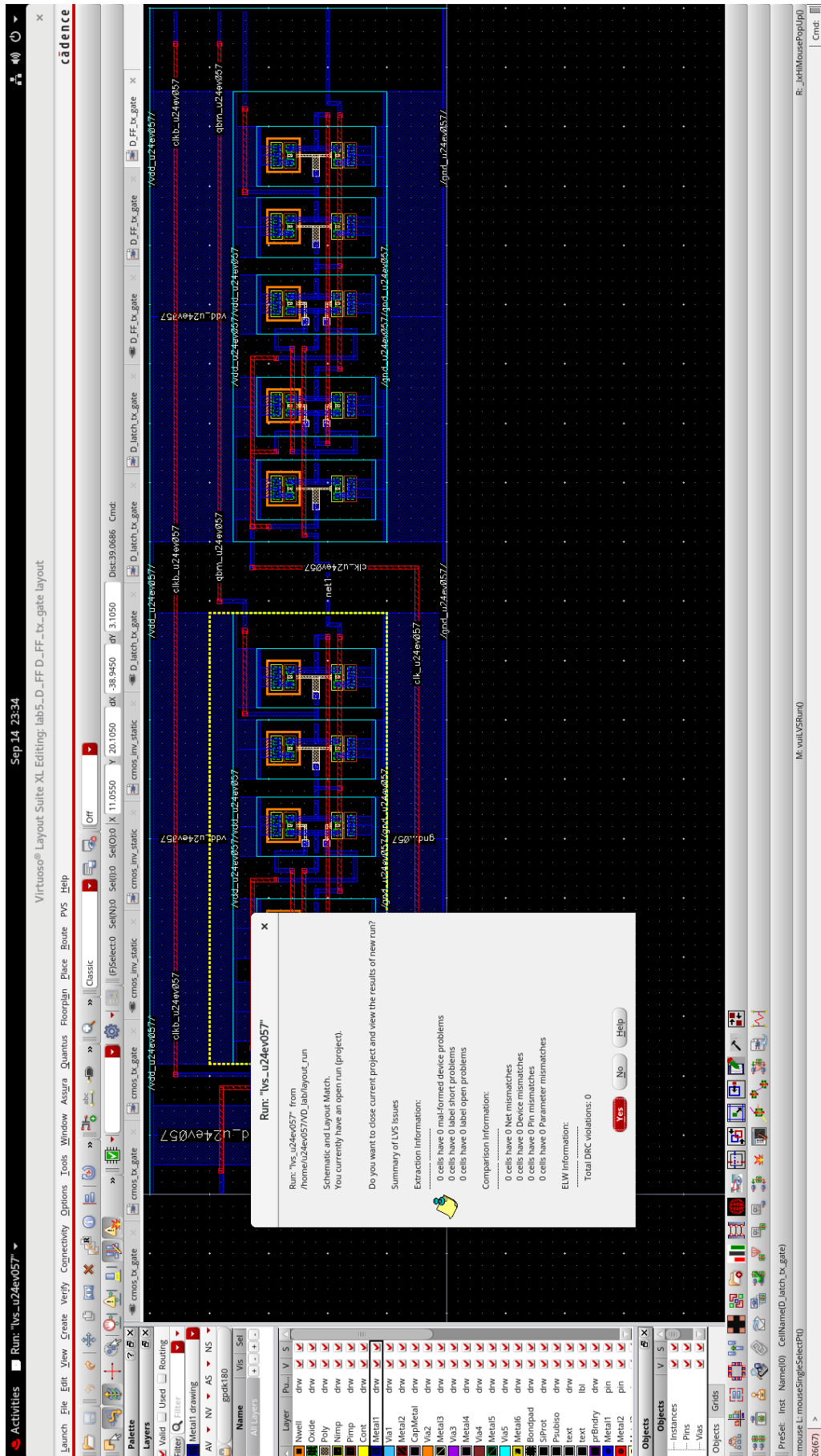


Figure 7.15: Tx Dff Lvs

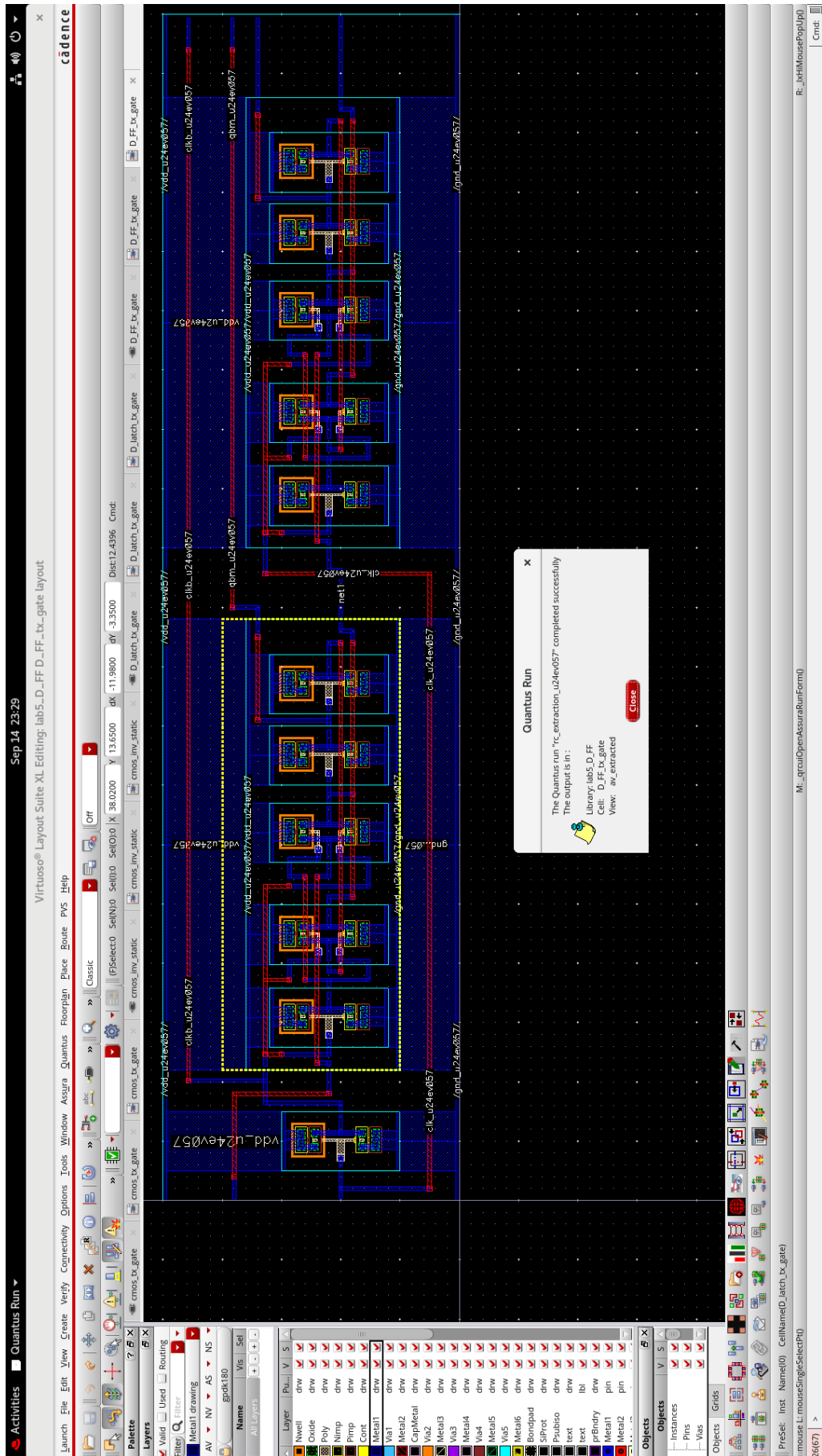


Figure 7.16: Tx Dff Quantus

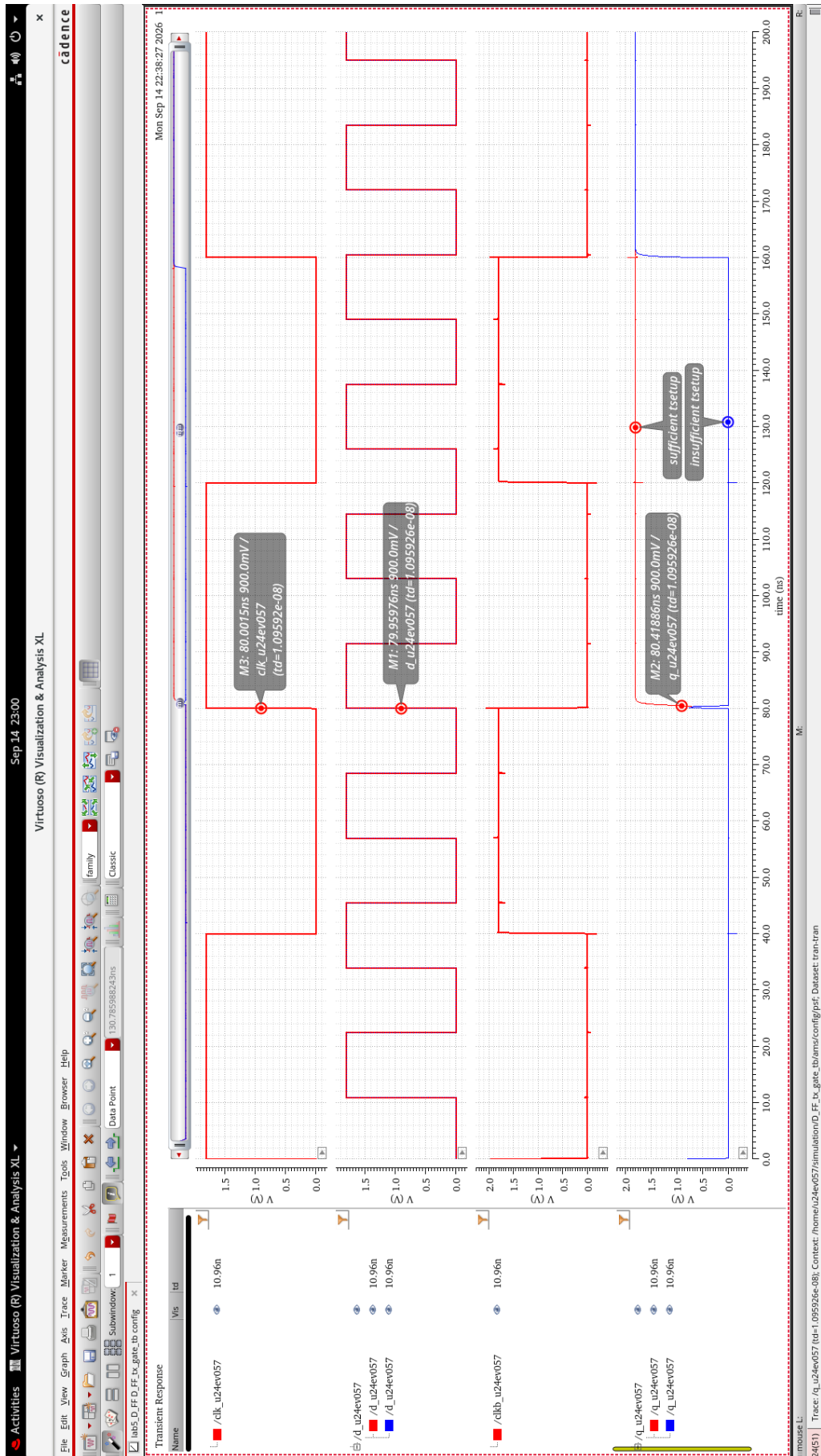


Figure 7.17: Tx Tran Pre Layout

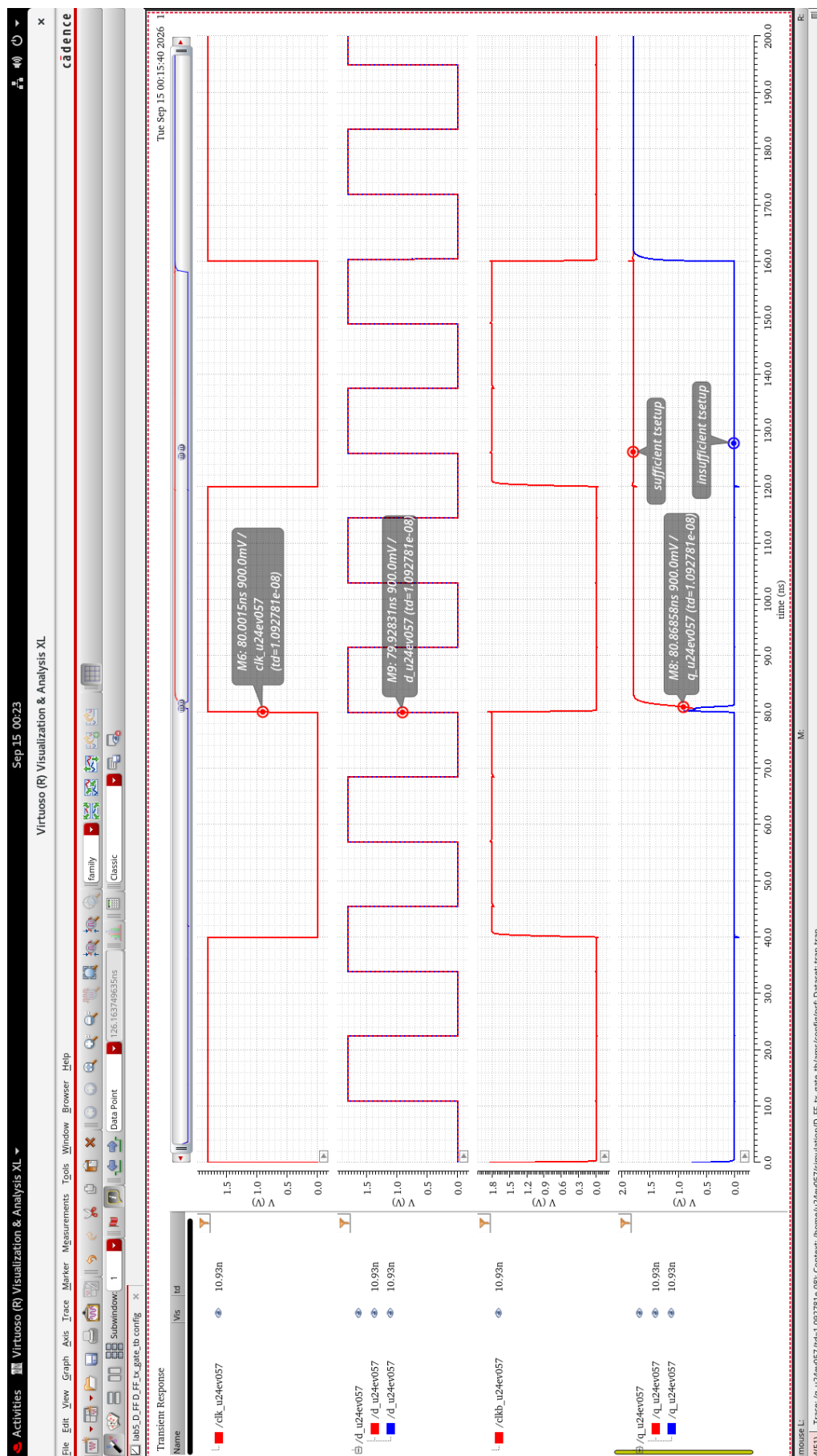


Figure 7.18: Tx Tran Post Layout

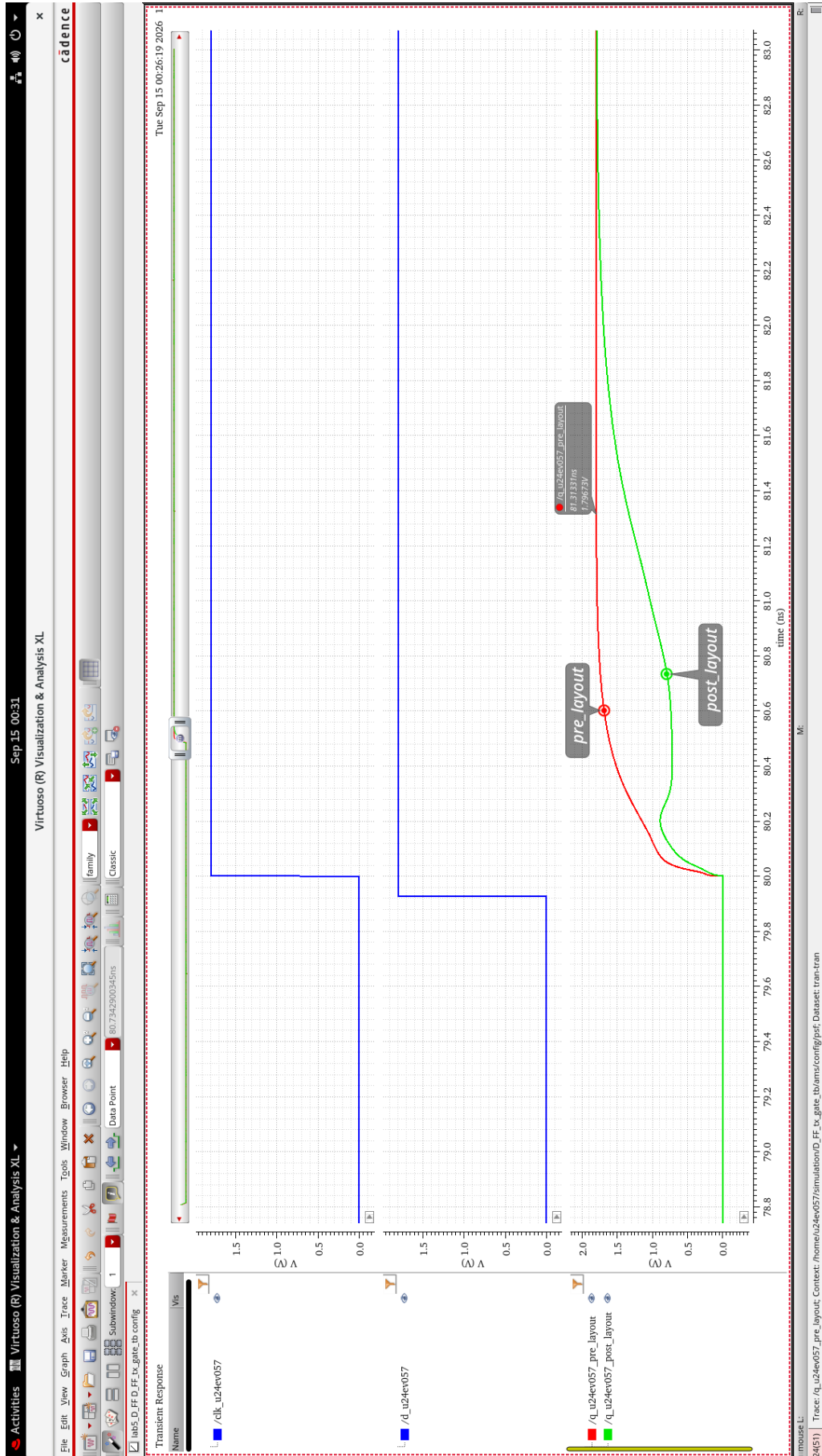


Figure 7.19: Tx Tran Pre Vs Post

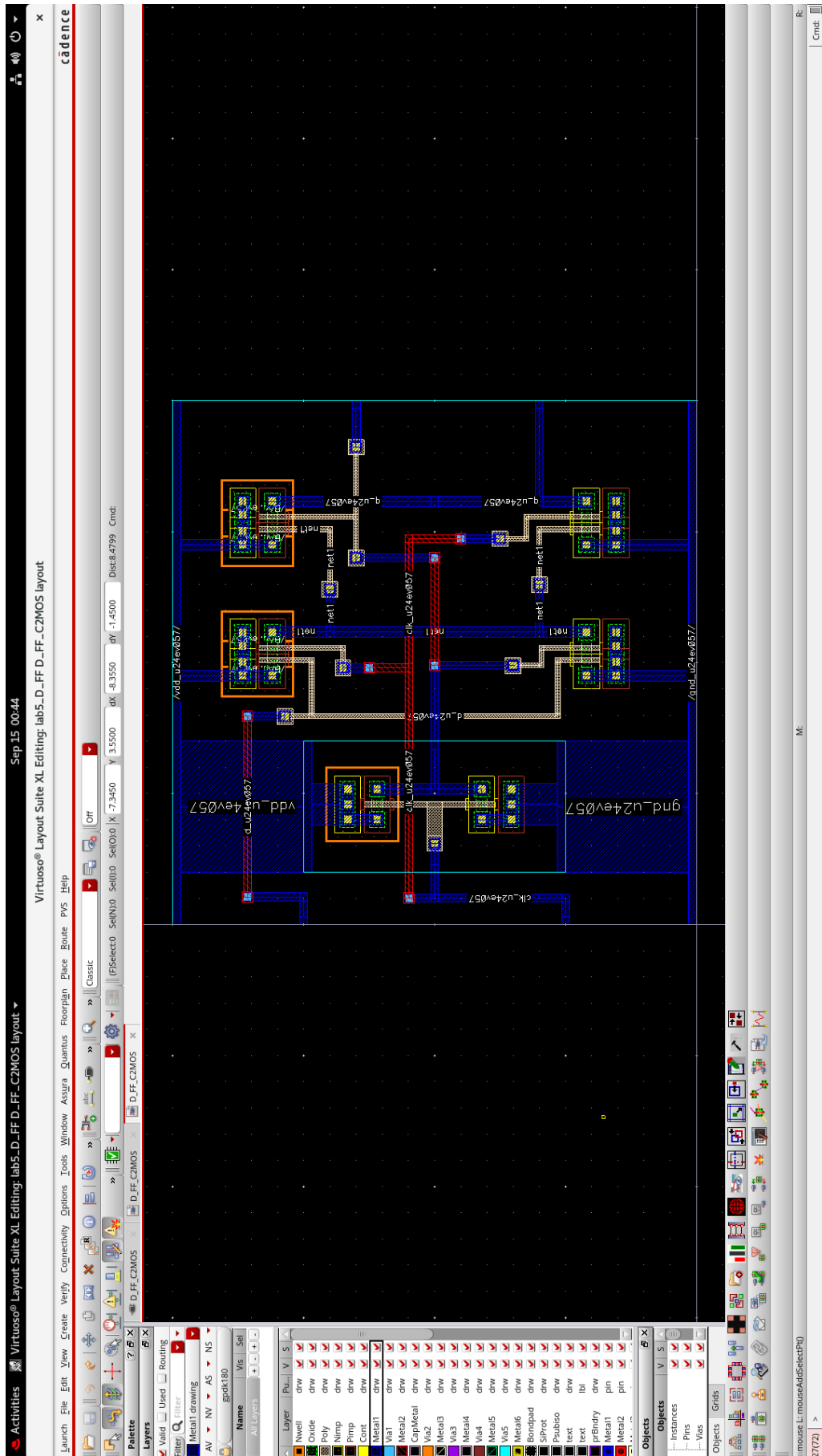


Figure 7.23: C2mos Dff Layout

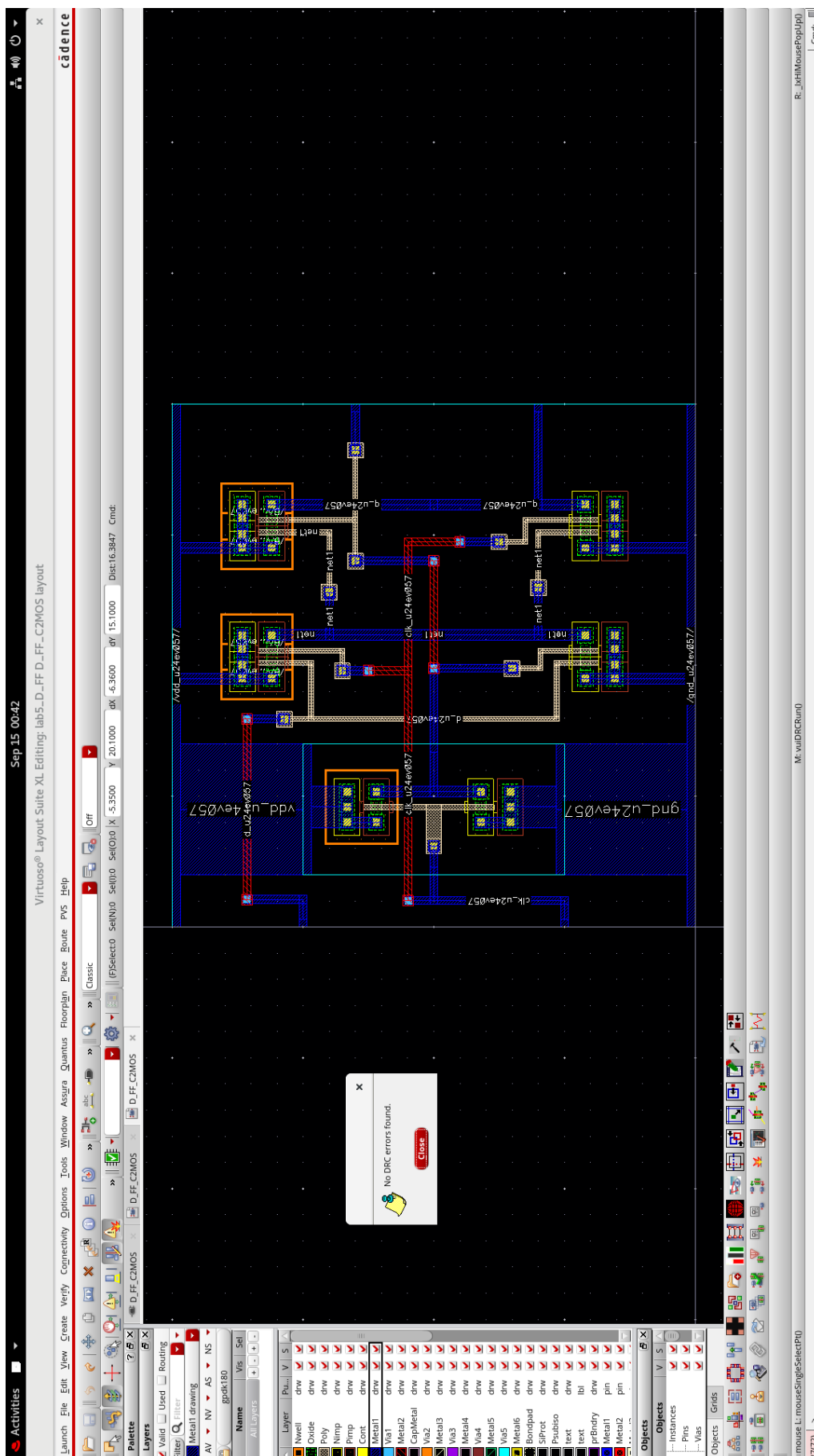


Figure 7.24: C2mos Dff Drc

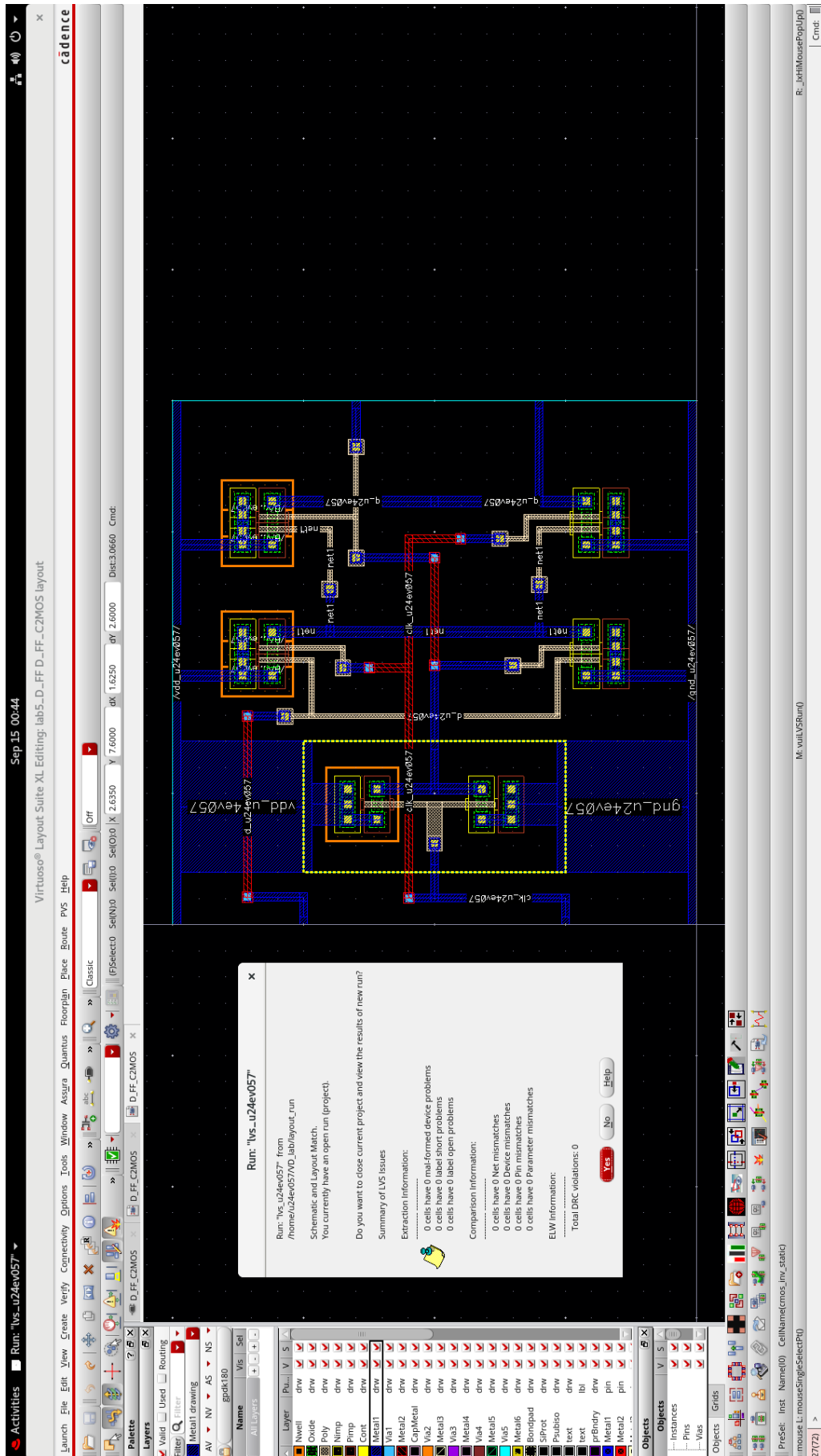


Figure 7.25: C2mos Dff Lvs

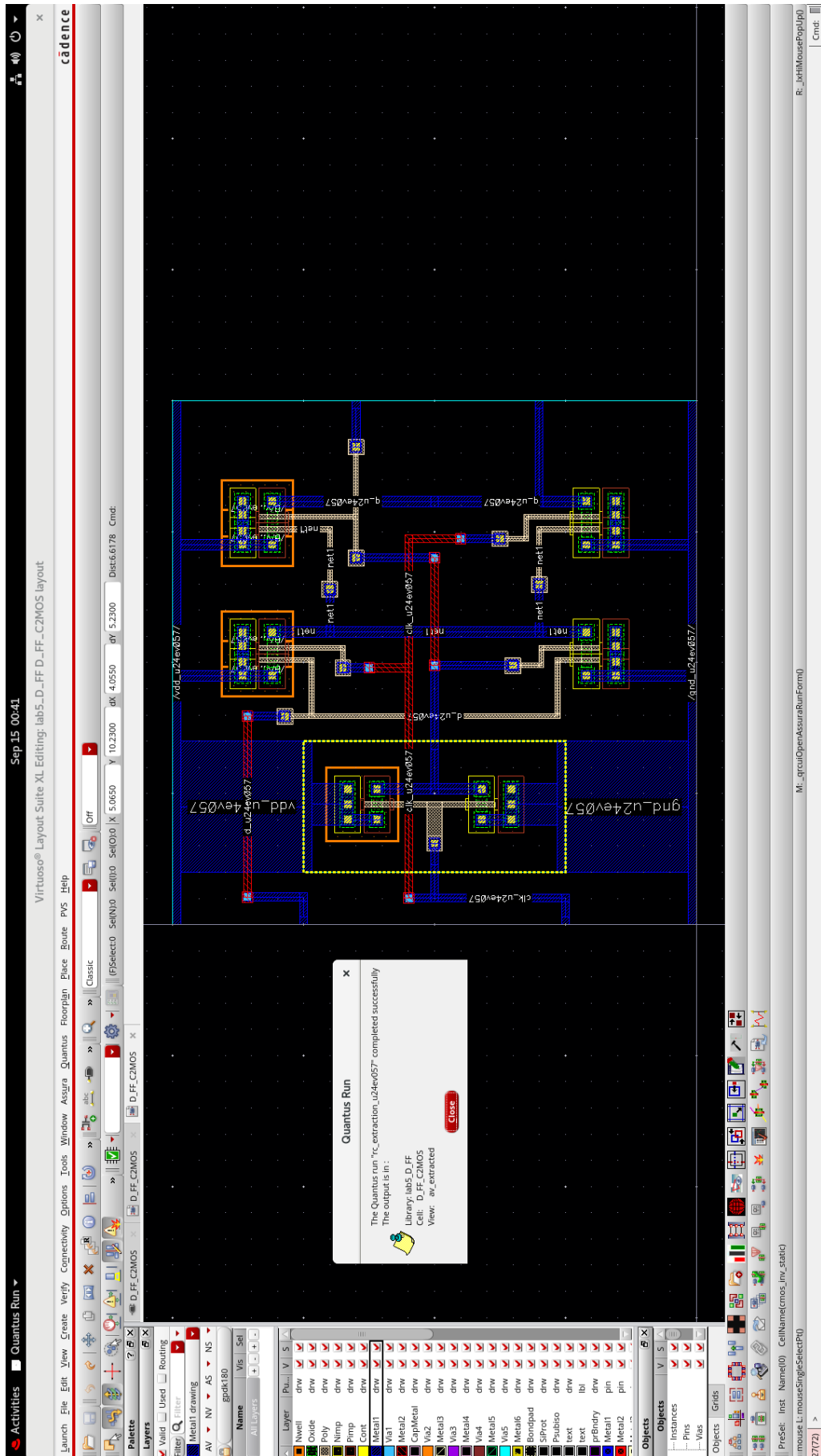


Figure 7.26: C2mos Dff Quantus

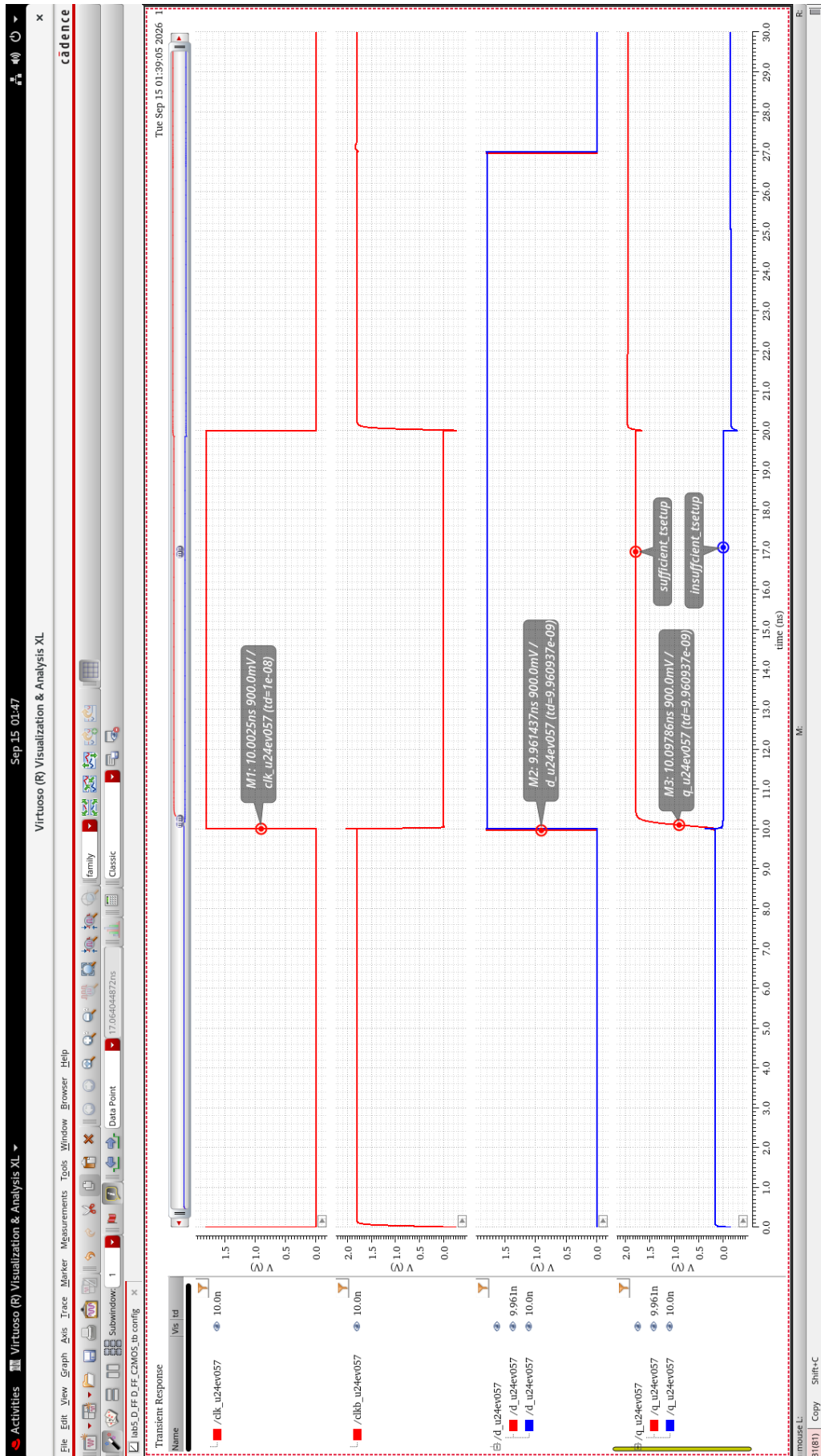


Figure 7.27: C2mos Dff Tran Pre Layout

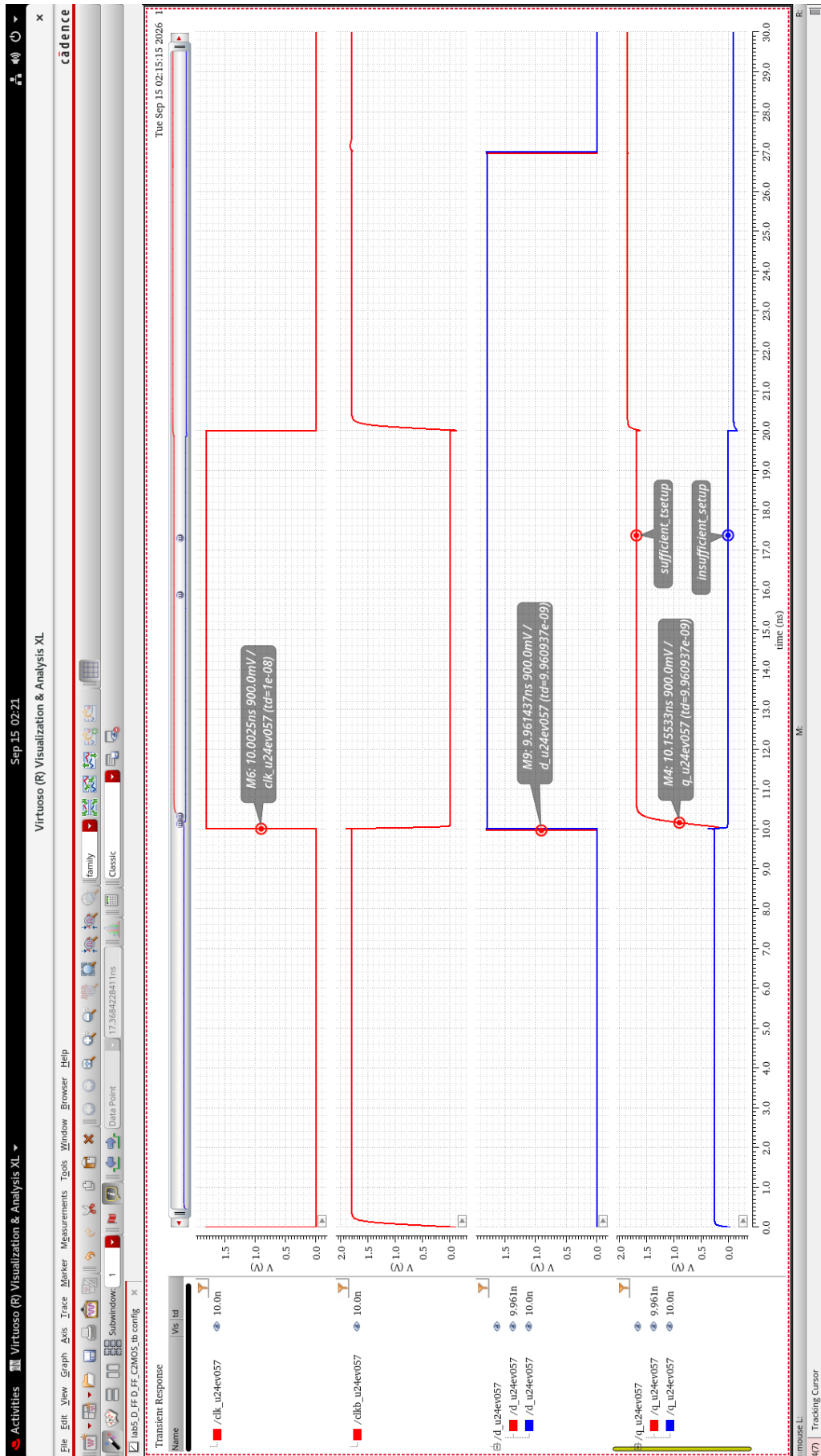


Figure 7.28: C2mos Tran Post Layout

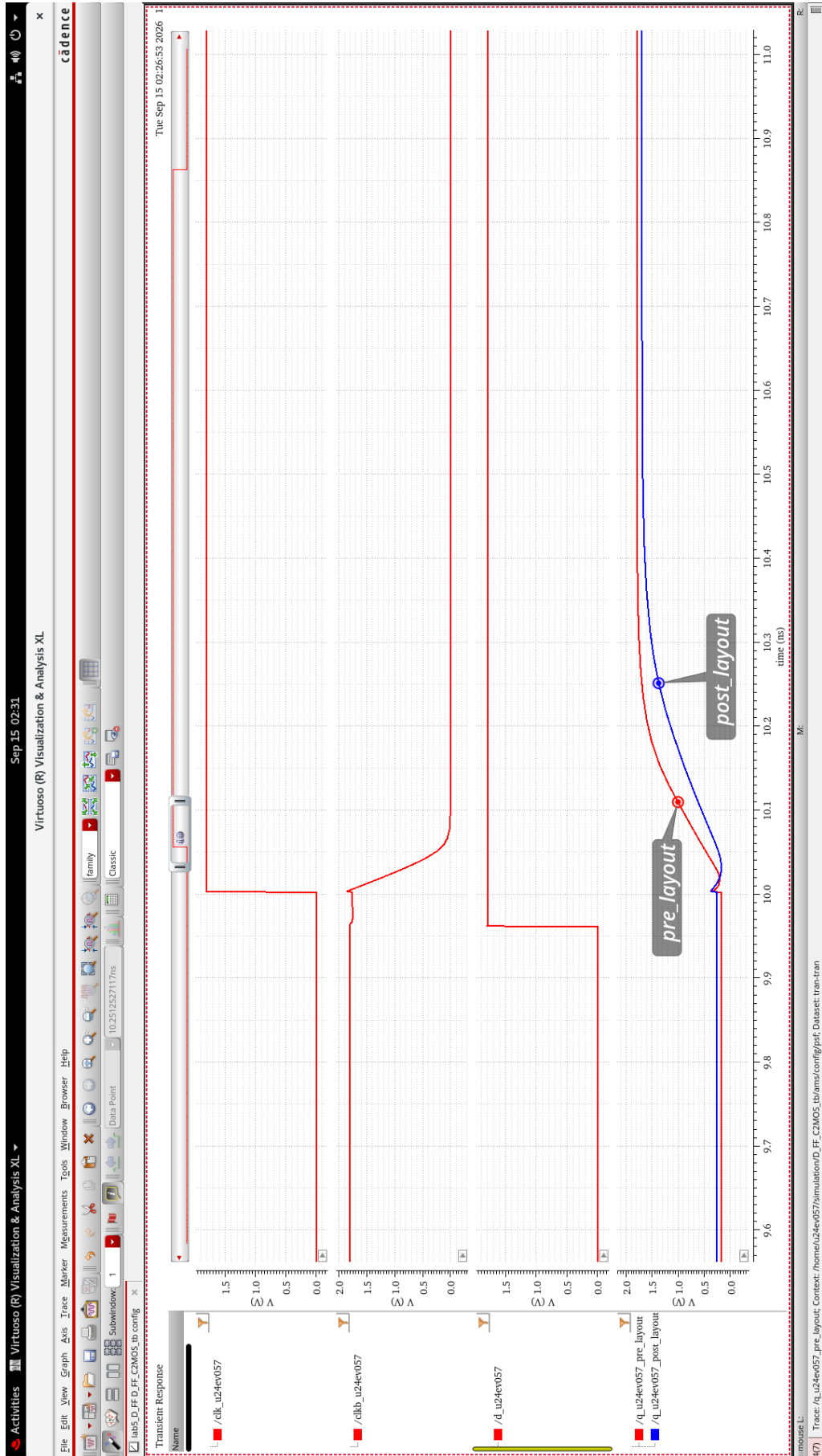


Figure 7.29: C2mos Tran Pre Vs Post

8. RESULT

The transmission-gate-based and C²MOS-based dynamic D flip-flops were successfully designed and verified. The pre-layout maximum frequencies were 2.178 GHz and 7.330 GHz, respectively, which reduced to 1.063 GHz and 5.157 GHz after considering layout parasitics.